Pranav Mohite

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Summary

Final-year B.Tech Computer Science (Cybersecurity) student with hands-on experience from three internships spanning web application security, vulnerability assessment, web development, and data analysis. Strong foundation in Python, SQL, Linux, networking, and security testing with Nmap and Burp Suite. Eager to apply emerging technologies, including AI, to build secure, reliable software in a fast-paced environment.

Education

Symbiosis Skills and Professional University Pune

B.Tech, Computer Science and Information Technology (Cybersecurity) — CGPA: 8.1 *Aug 2023 – June 2027*

Technical Skills

Programming Languages: C/C++ , Python, TypeScript, SQL, Motoko, Web Development
Cybersecurity: Vulnerability Assessment, Application Security, OWASP Top 10, Broken Access Control, Threat Modeling, Security Testing, Post-Quantum Cryptography (PQC)
Security Tools: Burp Suite, Nmap, Wireshark, Metasploit
Frameworks & Testing: Flask, React, pytest, Vitest, Unit Testing, Integration Testing, Test Automation
AI/ML: Scikit-learn, Supervised & Unsupervised Learning, Model Training & Evaluation
DevOps & Tools: Git, GitHub, GitHub Actions, CI/CD Pipelines, Docker, Claude Code
Systems & Networking: Linux, Windows, TCP/IP, DNS, Network Security, Access Control

Technical Projects

Ensemble ML Email Header Forensics Framework | *Python,Isolation Forest, SHAP*

- Developing an **ML-based email forensics engine** using **Random Forest, XGBoost, and Isolation Forest** to detect suspicious email-header patterns.
- Engineering features from **SPF, DKIM, DMARC, IP hops, timestamps, and routing metadata** to support suspicious-email detection.
- Applying **SHAP explainability** to provide investigator-facing insights into model predictions and suspicious email indicators.

QuantumMail (PQMail): Post-Quantum Secure Email Client | *Python, Flask, SQLite*

- Built a secure email client using hybrid **X25519 + ML-KEM-768 key exchange**, **ML-DSA-65 digital signatures**, **AES-256-GCM encryption**, and **TOTP-based 2FA**.
- Identified and remediated **24 security flaws**, including **CSRF, key hijacking, and path traversal**; implemented a **key transparency log** and a **14-attack security testing lab**.
- Authored **258 pytest unit tests** and extended a **BB84 quantum key distribution (QKD) simulator** with error correction.

OceanMind AI: Marine Risk Monitoring Platform | *React, TypeScript, Motoko, Vitest*

- Fixed **Broken Access Control (OWASP Top 10 A01)** in the Motoko backend, preventing unauthorized access to and modification of other users' records through **ownership validation checks**.
- Built a **249-test Vitest suite** achieving approximately **94% coverage** of scoring logic and integrated **GitHub Actions CI**; tests exposed **3 bugs**, including a scale mismatch that incorrectly classified low-risk data as Critical.
- Integrated real **NOAA and AIS data** and implemented **loitering, data-gap, and vessel-speed detection rules**, reducing flagged vessels from **120 to 81** in one day's dataset.

Flux: Automated Web Vulnerability Assessment System | *Python, Nmap*

- Built an automated **port scanner using Nmap** to identify exposed network services and support web security assessments.
- Developed an **OWASP-aligned HTTP Security Headers Analyzer** to detect missing security headers and common web configuration weaknesses.
- Generated automated **vulnerability assessment reports** with actionable remediation recommendations to improve application security and support **security hardening**.

Experience

Project-Based Data Analyst Intern Jun 2026 – Jul 2026

Oppidan India Group

- Cleaned, processed, and validated **business, sales, and survey datasets** using **Excel and Python** to improve data quality and consistency.
- Used **SQL queries and Excel functions** to organize records, identify data inconsistencies, and support business data analysis.
- Prepared datasets for analysis by handling missing values, removing duplicates, and standardizing data formats to ensure data accuracy.

Web Designing and Development Intern Jun 2025 – Aug 2025

Kalangan Arts

- Developed **responsive web pages** using **HTML5, CSS3, and JavaScript**, ensuring cross-browser compatibility and mobile responsiveness.
- Implemented **frontend UI components**, page layouts, and navigation elements to enhance usability and user experience.
- Performed **frontend debugging, layout optimization, and UI testing** to improve website functionality, consistency, and performance.

Cybersecurity and Web Development Intern Jun 2024 – Aug 2024

Technomind Machine Automation

- Conducted **web application security assessments** using **Nmap and Burp Suite** to identify potential vulnerabilities and security misconfigurations.
- Recommended **security remediation measures**, tracked identified issues through resolution, and supported web development activities alongside security testing.
- Developed **SmartStock**, an automated **inventory management and procurement system** to support tool inventory tracking and procurement workflows.

Achievements

Indradhanu Grand Challenge (PCCOE): Ranked **Top 5 among 1,000+ teams** and received the **Special Category Award (INR 20,000)** for OceanMind AI.

Smart India Hackathon 2025: Recognized among the **Top 10 University Teams**.

Certifications and Languages

Certifications: Oracle Agentic AI Foundations Associate (1Z0-1157-26); Saylor Academy – Information Security, Networks, and Databases.

Languages: English, Hindi, Marathi, German (**Basic Proficiency**).